```python
# Install OpenCV (if needed)
!pip install opencv-python

# Import libraries
import cv2
import matplotlib.pyplot as plt
from google.colab import files

# Upload image
uploaded = files.upload()

# Get uploaded file name
image_path = list(uploaded.keys())[0]

# Read the image
image = cv2.imread(image_path)

# Check if image is loaded
if image is None:
    print("Error: Unable to load image.")
else:
    # Convert BGR to RGB
    image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

    # Convert to Grayscale
    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

    # Apply Gaussian Blur to reduce noise
    blurred = cv2.GaussianBlur(gray, (5, 5), 0)

    # Perform Canny Edge Detection
    edges = cv2.Canny(blurred, threshold1=100, threshold2=200)

    # Display Original and Edge Images
    plt.figure(figsize=(12,6))

    plt.subplot(1,2,1)
    plt.imshow(image_rgb)
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    plt.title("Original Image")
    plt.axis("off")

    plt.subplot(1,2,2)
    plt.imshow(edges, cmap='gray')
    plt.title("Canny Edge Detection")
    plt.axis("off")

    plt.show()

    # Save the output image
    output_path = "canny_edge_output.png"
    cv2.imwrite(output_path, edges)

    print("Edge detection completed successfully!")
```

```
Requirement already satisfied: opencv-python in /usr/local/lib/python3.12/dist-packages (5.0.0.93)
Requirement already satisfied: numpy>=2 in /usr/local/lib/python3.12/dist-packages (from opencv-python) (2.0.2)
```

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Saving istockphoto-1598719537-612x612.jpg to istockphoto-1598719537-612x612.jpg
```

Original Image

Canny Edge Detection